

BuyWise Overview

BuyWise is a link-first buying assistant for people checking used listings and retail deals before they pay. The core product promise is simple: paste a product link, and BuyWise tells the buyer what it can verify, what still needs confirmation, and whether the deal looks safe enough to keep checking.

This document reflects the current product state as of May 13, 2026. It is intentionally honest about what works, what is live, what requires API credentials, and what is still limited.

Current Website Structure

The homepage is the main analyzer entry point. A user can paste a resale or retail product link and BuyWise tries to read the page. The top navigation includes Search, Analyze Listing, Saved, and Log in. About Us is available in the footer.

/search is restored as a real live-offer search page. It no longer redirects home. It has a product search bar, source filter, condition filter, max price filter, loading state, empty state, provider status cards, and live offer cards.

/submit is the full analyzer page. It receives drafts from the homepage and can auto-run when enough information is present. If a price cannot be read confidently, BuyWise asks the user to confirm it before scoring.

/saved lets logged-in users view saved verdicts and saved offers. Supabase sync works when the user is authenticated. Logged-out users still have local saved-item fallback.

/auth supports account signup, login, and logout through Supabase Auth.

/about is customer-facing and explains what BuyWise does. It is not meant to show internal strategy or MVP notes.

What The Analyzer Does Today

The analyzer accepts a product or listing URL. It normalizes the URL, identifies the source, infers whether the link is resale or retail, and tries to extract useful product data.

The analyzer can return a verdict page with a deal score, market position label, risk level, confidence level, price confidence, suggested offer guidance, red flags, positive trust signals, buyer questions, better resale moves, better retail moves, and data-source notes.

The analyzer is strict about low-confidence prices. If BuyWise sees a possible price in visible page text but the surrounding context looks weak, unrelated, too small, or possibly shipping/financing/accessory text, it does not use that value as the final price. The user must confirm the price before scoring.

The analyzer no longer force-matches unclear products to preset guide records. For example, a MacBook link should not silently become a Sony camera benchmark. If BuyWise cannot verify an exact market benchmark, it scores from the extracted and user-confirmed link details only.

Live Offer Search

Search now uses a live-offer pipeline instead of preset sample products. Product cards are only shown when a configured provider returns real offer data.

Each live offer card can show product title, source, price, condition, image, source method, confidence, price confidence, warning text, original offer link, and a save button.

If no live provider credentials are configured, Search does not pretend. It shows that eBay, Best Buy, Amazon, and Walmart providers are missing configuration and explains that API credentials are required before real product cards can appear.

This means the current local or production site may show zero Search results until API keys are added. That is expected and better than showing fake products.

Current Data Pipeline

The intended hierarchy is now represented in code:

1. Official marketplace or retailer APIs
2. Affiliate/product data APIs as future provider adapters
3. Structured page metadata extraction
4. Server-side rendered HTML parsing
5. Optional headless browser fallback in the future
6. Manual fallback only for missing fields

The current implemented provider adapters are eBay Browse API, Best Buy Product API, Amazon Product Advertising API, and Walmart catalog/search access through configured credentials and endpoint. These adapters do not return fake data when credentials are missing.

The existing `/api/extract-link` route now tries provider lookup first for supported source links. If a provider is missing, fails, or does not return enough data, it falls back to public page metadata and server-rendered HTML parsing.

Provider Credentials Needed

For strong live data, these credentials need to be added locally and in Vercel:

EBAY_CLIENT_ID and EBAY_CLIENT_SECRET for eBay Browse API.

BEST_BUY_API_KEY for Best Buy Product API.

AMAZON_PAAPI_ACCESS_KEY, AMAZON_PAAPI_SECRET_KEY, and AMAZON_ASSOCIATE_TAG for Amazon Product Advertising API.

WALMART_SEARCH_ENDPOINT plus compatible Walmart catalog/API credentials for Walmart data.

Without these keys, BuyWise can still read some public page metadata from pasted links, but it cannot reliably perform broad live product search or guaranteed price extraction from blocked or JavaScript-heavy pages.

Why Some Links Still Do Not Extract Well

Many major retail and marketplace pages do not expose the true buy-box price in static HTML. Prices may be rendered client-side, personalized by location/account, hidden behind internal APIs, blocked by anti-bot systems, or only available through official/affiliate APIs.

Amazon is a good example. Public HTML can include unrelated prices such as accessory prices, financing amounts, shipping numbers, coupons, crossed-out prices, or sponsored product prices. BuyWise should not trust those numbers unless the context is strong or the official Amazon API confirms the offer.

eBay item pages may block normal server fetches, but eBay Browse API can provide reliable listing data when credentials are configured.

Best Buy, Walmart, Target, and similar retailers usually need official or partner APIs for reliable search and pricing. Public page parsing is a fallback, not the main strategy.

Facebook Marketplace is not fetched. Users should paste listing details manually for Facebook links. Screenshot upload or a browser extension would be better long-term approaches.

Scoring And Trust Model

BuyWise separates price attractiveness from trust and confidence. A low price should not automatically produce a high score. Very low prices can increase scam suspicion if proof is weak.

The score considers price confidence, extraction confidence, trust signals, risk terms, condition wording, marketplace/source type, listing completeness, warranty/return clues, and whether the price had to be manually confirmed.

If price confidence is under the scoring threshold, the analyzer blocks scoring and asks the user to enter the actual price. This prevents fake 90+ scores from bad extraction.

The verdict is intentionally cautious when BuyWise lacks reliable price or product identity. This is a trust feature, not a bug.

Saving And Accounts

Supabase Auth is used for accounts. Users can sign up, log in, log out, and save analyzed verdicts or live offer cards.

When logged in, saved items are inserted into the saved_items table with the authenticated user_id. Row-level security should allow users to select, insert, update, and delete only their own rows.

When not logged in, BuyWise stores saved items locally on the device. This keeps the app usable even without an account.

Known Limitations

Live Search depends on provider credentials. If no keys are configured, Search will show provider setup states instead of product cards.

Universal extraction from arbitrary links is not realistic. Some links will still need manual price or product confirmation.

Headless browser rendering is not enabled by default. It should be added only behind a strict flag because it increases hosting complexity and still cannot bypass CAPTCHAs or private content.

BuyWise does not currently scrape Facebook Marketplace.

BuyWise does not yet have screenshot upload, OCR, browser extension support, or full real-time market trend history.

The product is moving away from static guides. Any remaining internal benchmark logic should be treated as support infrastructure, not the main user-facing concept.

What Changed In This Pass

Search was restored as a real page.

A new `/api/search-offers` endpoint was added for live provider search.

A live offer provider service was added with adapters for eBay, Best Buy, Amazon PA-API, and Walmart-compatible catalog access.

The analyzer now attempts provider lookup before falling back to metadata and HTML parsing.

The UI now refuses to show fake product cards when provider keys are missing.

The README and environment example now document the live provider setup honestly.

This PDF was updated to reflect the actual state of the product rather than implying that all live link analysis works perfectly without credentials.